Senzing Data Discoveries

What Senzing found in your data

Evidence source: data/query/master_list.jsonl, generated by the Java query.QueryResults SDK export using SZ_EXPORT_DEFAULT_FLAGS; per-record and relationship rows below are from that export. Worked Why, How, and network examples are from the SDK-backed programs in src/query/.

Scope note: This is a descriptive review of the loaded data, not an accuracy verdict. The checkout has no actual_truthset_key.csv answer key, so precision, recall, F1, and absolute match accuracy are unmeasured.

Headline numbers, interpreted

- **Records loaded:** 159 (AML_WATCHLIST: 17, BANK_CUSTOMERS: 120, BUSINESS_REGISTRY: 22). These are source records, not distinct people or organizations.
- **Resolved entities:** 120. The SDK grouped the loaded records into these entity identities.
- **Net consolidation:** 39 records have non-empty per-record match keys beyond the first record in each resolved entity; 120 entities remain. This is a count of the extra records represented in merged entities, not a count of independent business decisions.
- **Multi-record entities:** 30; entity sizes are 90 single record, 24 2 records, 4 3 records, 1 4 records, 1 5 records.
- **Cross-source resolved entities:** 9/120 (7.5%). Observed source-pair overlap: AML_WATCHLIST/BANK_CUSTOMERS: 1, BANK_CUSTOMERS/BUSINESS_REGISTRY: 8.
- **Distinct relationship links:** 153, deduplicated by the unordered entity-ID pair because the export reports a relationship from both endpoints. Categories: 49 POSSIBLY_SAME, 90 POSSIBLY_RELATED, 14 disclosed, and 2 ambiguous links.

Data source	Records	Resolved entities containing source	Source compression
AML_WATCHLIST	17	16	5.9%
BANK_CUSTOMERS	120	91	24.2%
BUSINESS_REGISTRY	22	22	0.0%

Source entity counts overlap when one resolved entity contains records from multiple sources. Compression is descriptive; no source-profiler uniqueness baseline is available here to diagnose over- or under-matching from these percentages alone.

Merges and match keys

The table lists every non-empty per-record match key in the SDK export (39 rows). Rows are ordered by entity, source, and record ID; this is not a chronological sequence. Rule codes are reproduced as returned and are not expanded.

Entity	Data source	Record ID	Match level	MATCH_KEY	ERRULE_CODE
20000	AML_WATCHLIST	1012	RESOLVED	+NAME+DOB+ADDRESS	CNAME_CFF_CEXCL
5					
20000	BANK_CUSTOMERS	1009	RESOLVED	+DOB+SSN+PNAME	SF1_PNAME_CSTAB
5					

Entity	Data source	Record ID	Match level	MATCH_KEY	ERRULE_CODE
200005	BANK_CUSTOMERS	1010	RESOLVED	+NAME+DOB+ADDRESS	CNAME_CFF_CEXCL
200005	BANK_CUSTOMERS	1011	RESOLVED	+NAME+DOB+ADDRESS+PHONE	MFF_LNAME_CEXCL
400002	BANK_CUSTOMERS	2072	RESOLVED	+ADDRESS+PHONE+LIKELY_NAME	MFF_LNAME_CEXCL
400004	BANK_CUSTOMERS	1071	RESOLVED	+NAME+DOB	SNAME_SSTAB
400004	BANK_CUSTOMERS	1072	RESOLVED	+NAME+DOB+ADDRESS+NATIONAL_ID	SF1_SNAME_CFF_CSTAB
400005	BANK_CUSTOMERS	1069	RESOLVED	+NAME+DOB	SNAME_SSTAB
400005	BANK_CUSTOMERS	1070	RESOLVED	+NAME+DOB+ADDRESS+NATIONAL_ID	CF1_PNAME_CFF_CSTAB
400010	BANK_CUSTOMERS	2142	RESOLVED	+NAME+ADDRESS	CNAME_CFF
500001	BANK_CUSTOMERS	1075	RESOLVED	+NAME+DOB+ADDRESS	CNAME_CFF_CEXCL
500006	BANK_CUSTOMERS	1082	RESOLVED	+NAME+DOB+ADDRESS+PASSPORT	SF1_SNAME_CFF_CSTAB
500009	BUSINESS_REGISTRY	2074	RESOLVED	+NAME+ADDRESS	CNAME_CFF
600004	BANK_CUSTOMERS	2011	RESOLVED	+NAME+ADDRESS	CNAME_CFF
600006	BANK_CUSTOMERS	2042	RESOLVED	+NAME+ADDRESS	CNAME_CFF
600012	BANK_CUSTOMERS	2152	RESOLVED	+NAME+PHONE	CNAME_CFF
700001	BANK_CUSTOMERS	1002	RESOLVED	+NAME+DOB+ADDRESS+PHONE	MFF_LNAME_CEXCL
700001	BANK_CUSTOMERS	1003	RESOLVED	+NAME+DOB+ADDRESS+EMAIL	SF1_PNAME_CFF_CSTAB
700001	BANK_CUSTOMERS	1004	RESOLVED	+NAME+DOB+ADDRESS	CNAME_CFF_CEXCL
700005	BANK_CUSTOMERS	1088	RESOLVED	+NAME+DOB+PHONE+EMAIL-GENDER	SF1_SNAME_CFF_CSTAB_DEXCL
700008	BANK_CUSTOMERS	1031	RESOLVED	+NAME+DOB+ADDRESS	CNAME_CFF_CEXCL
700009	BANK_CUSTOMERS	1092	RESOLVED	+NAME+DOB+ADDRESS+PHONE+EMAIL	SF1_SNAME_CFF_CSTAB
700017	BANK_CUSTOMERS	1023	RESOLVED	+NAME+DOB	SNAME_SSTAB
700020	BANK_CUSTOMERS	1077	RESOLVED	+NAME+DOB+ADDRESS	CNAME_CFF_CEXCL
700022	BANK_CUSTOMERS	1063	RESOLVED	+NAME+DOB	SNAME_SSTAB
700023	BANK_CUSTOMERS	1079	RESOLVED	+NAME+DOB+ADDRESS+PHONE	MFF_LNAME_CEXCL
700029	BANK_CUSTOMERS	1083	RESOLVED	+NAME+DOB+ADDRESS+PASSPORT	SF1_PNAME_CFF_CSTAB
700030	BANK_CUSTOMERS	1034	RESOLVED	+NAME+DOB+ADDRESS	CNAME_CFF_CEXCL
700030	BANK_CUSTOMERS	1035	RESOLVED	+NAME+DOB+ADDRESS	CNAME_CFF_CEXCL
700031	BANK_CUSTOMERS	1094	RESOLVED	+DOB+DRLIC+PNAME	SF1_PNAME_CSTAB

Entity	Data source	Record ID	Match level	MATCH_KEY	ERRULE_CODE
700032	BANK_CUSTOMERS	2193	RESOLVED	+NAME+DOB+ADDRESS	CNAME_CFF_CEXCL
700035	BANK_CUSTOMERS	1050	RESOLVED	+NAME+DOB+PASSPORT	SF1_PNAME_CSTAB
700044	BANK_CUSTOMERS	1054	RESOLVED	+DOB+ADDRESS+DRLIC+PNAME	SF1_PNAME_CFF_CSTAB
700049	BANK_CUSTOMERS	1033	RESOLVED	+NAME+DOB+PHONE	CNAME_CFF_CEXCL
700050	BANK_CUSTOMERS	1015	RESOLVED	+NAME+DOB+ADDRESS	CNAME_CFF_CEXCL
700050	BANK_CUSTOMERS	1018	RESOLVED	+NAME+DOB+PHONE	CNAME_CFF_CEXCL
700056	BANK_CUSTOMERS	1095	RESOLVED	+NAME+DOB+DRLIC	SF1_PNAME_CSTAB
700076	BANK_CUSTOMERS	1074	RESOLVED	+NAME+DOB+ADDRESS	CNAME_CFF_CEXCL
700092	BANK_CUSTOMERS	1085	RESOLVED	+NAME+DOB+PHONE+EMAIL	SF1_PNAME_CFF_CSTAB

Match-key frequency across those records:

MATCH_KEY	Count
+NAME+DOB+ADDRESS	11
+NAME+ADDRESS	4
+NAME+DOB	4
+NAME+DOB+ADDRESS+PHONE	3
+NAME+DOB+ADDRESS+NATIONAL_ID	2
+NAME+DOB+ADDRESS+PASSPORT	2
+NAME+DOB+PHONE	2
+ADDRESS+PHONE+LIKELY_NAME	1
+DOB+ADDRESS+DRLIC+PNAME	1
+DOB+DRLIC+PNAME	1
+DOB+SSN+PNAME	1
+NAME+DOB+ADDRESS+EMAIL	1
+NAME+DOB+ADDRESS+PHONE+EMAIL	1
+NAME+DOB+DRLIC	1
+NAME+DOB+PASSPORT	1
+NAME+DOB+PHONE+EMAIL	1
+NAME+DOB+PHONE+EMAIL-GENDER	1
+NAME+PHONE	1

Review queue

These are the 15 deduplicated cross-source POSSIBLY_SAME or ambiguous pairs from the SDK-exported RELATED_ENTITIES. They are review candidates, not merges or verdicts. The ambiguous flag is reported separately from the match level.

Entity A	Source(s) A	Entity B	Source(s) B	Match level	Ambiguous	MATCH_KEY
200001	AML_WATCHLIST	700001	BANK_CUSTOMERS	POSSIBLY_SAME	no	+NAME+ADDRESS-DOB
200001	AML_WATCHLIST	700042	BANK_CUSTOMERS	POSSIBLY_SAME	no	+NAME+DRLIC-DOB
200002	AML_WATCHLIST	500008	BANK_CUSTOMERS	POSSIBLY_SAME	no	+NAME+ADDRESS-GENERATION-DOB
200003	AML_WATCHLIST	600007	BUSINESS_REGISTRY	POSSIBLY_SAME	no	+NAME+ADDRESS+PHONE-DOB

Entity A	Source(s) A	Entity B	Source(s) B	Match level	Ambiguous	MATCH_KEY
200004	AML_WATCHLIST	700038	BANK_CUSTOMERS	POSSIBLY_SAME	no	+NAME+ADDRESS-DOB
200004	AML_WATCHLIST	700104	BANK_CUSTOMERS	POSSIBLY_SAME	no	+NAME+ADDRESS-DOB
200007	AML_WATCHLIST	700033	BANK_CUSTOMERS	POSSIBLY_SAME	no	+ADDRESS+SSN-DOB
200008	AML_WATCHLIST	600011	BUSINESS_REGISTRY	POSSIBLY_SAME	no	+NAME+EMPLOYER-DOB
200010	AML_WATCHLIST	700038	BANK_CUSTOMERS	POSSIBLY_SAME	no	+NAME+ADDRESS-DOB
200010	AML_WATCHLIST	700104	BANK_CUSTOMERS	POSSIBLY_SAME	no	+NAME+ADDRESS-DOB-GENDER
300002	AML_WATCHLIST	600008	BUSINESS_REGISTRY	POSSIBLY_SAME	no	+NAME+ADDRESS+PHONE-DOB
300002	AML_WATCHLIST	700006	BANK_CUSTOMERS	POSSIBLY_SAME	no	+NAME+PHONE-DOB
300003	AML_WATCHLIST	700030	BANK_CUSTOMERS	POSSIBLY_SAME	yes	+PNAME+DOB+EMAIL (Ambiguous)
300006	AML_WATCHLIST	700030	BANK_CUSTOMERS	POSSIBLY_SAME	yes	+PNAME+DOB+EMAIL (Ambiguous)
300007	AML_WATCHLIST	400009	BUSINESS_REGISTRY	POSSIBLY_SAME	no	+NAME+EMPLOYER-DOB

Why and how: worked examples

Resolved-entity record evidence: getEntity returned Entity 200005 (Eddie Kusha) with five records. The record data shows the following names and birth dates: AML_WATCHLIST:1014, Eddie Knight, 3/1/70; AML_WATCHLIST:1012, Eddie Kusha, 3/1/1970; BANK_CUSTOMERS:1009, Edward Kusha, 1970-03-01; BANK_CUSTOMERS:1011, Ed Knight, 03/01/70; and BANK_CUSTOMERS:1010, Eddie Kusha, 1970-03-01. These are the actual values behind the Why and How outputs below; no SSN or other direct identifier is reproduced here.

Why example: AML_WATCHLIST:1014 and BANK_CUSTOMERS:1009 both resolve to Entity 200005 (MATCH_LEVEL_CODE=RESOLVED). The returned WHY_KEY is +PNAME+DOB+SSN, and the reported rule is SF1_PNAME_CSTAB (left as an internal code). WHY_KEY_DETAILS.CONFIRMATIONS reports NAME at 71/PLAUSIBLE from /PART_NAME, DOB at 100/SAME, and SSN at 100/SAME. The accompanying feature scores include ADDRESS 42/NO_CHANCE, PHONE 0/NO_CHANCE, and RECORD_TYPE 100/SAME. Here the numeric scores use the returned score values and SCORE_BUCKET labels to describe each comparison.

How example: Entity 200005 contains five records and has four resolution steps. The SDK reported NEED_REEVALUATION=0 and one final virtual entity containing all five records.

- Step 1: AML_WATCHLIST:1014 joined BANK_CUSTOMERS:1009 with +PNAME+DOB+SSN; confirmations were NAME 71/PLAUSIBLE, DOB 100/SAME, and SSN 100/SAME.
- Step 2: the two-record virtual entity joined BANK_CUSTOMERS:1010 with +NAME+DOB+ADDRESS; confirmations were NAME 97/CLOSE, DOB 100/SAME, and ADDRESS 100/SAME.
- Step 3: that virtual entity joined BANK_CUSTOMERS:1011 with +NAME+DOB+ADDRESS+PHONE;

confirmations were NAME 97/CLOSE, DOB 100/SAME, ADDRESS 95/CLOSE, and PHONE 90/CLOSE.

- Step 4: that virtual entity joined AML_WATCHLIST:1012 with +NAME+DOB+ADDRESS; confirmations were NAME, DOB, and ADDRESS at 100/SAME.

Near-miss (the one that teaches more):

Entity 300003 is Maria Medina; its record AML_WATCHLIST:1038 has NAME_FIRST=Maria, NAME_LAST=Medina, DATE_OF_BIRTH=12/11/73, and address Watchlist Demo City, NV. Entity 700030 is Maria Luis Medina Sentosa; its records are BANK_CUSTOMERS:1036 (Maria Luis Sentosa, 11/12/73), 1035 (Maria Luis M Sentosa, 11/12/73), and 1034 (Maria Luis Medina Sentosa, 11/21/73), all at Customer Demo City, CA. The SDK reports POSSIBLY_SAME with IS_AMBIGUOUS=1 and MATCH_KEY=+PNAME+DOB+EMAIL (Ambiguous); Why details report NAME/LIKELY, DOB/CLOSE, and EMAIL/SAME. The different observed names and dates are concrete review evidence, not proof either way. These entities remain separate, so treat this as a human-review candidate, not a resolved merge.

Relationship networks

The SDK findNetwork query was seeded with Entity 200004 (John Smith), Entity 200005 (Eddie Kusha), and Entity 200010 (John Smith), the highest-relationship candidates from step 4a. It returned 21 entities and 62 links. Entity 200004 and Entity 200005 each had 11 relationships; Entity 200010 had 10.

Using Entity 200004 as the hub, Entities 200001 and 200010 were two neighbors with no direct link between them. findPath confirmed this shortest path:

```
Entity 200001 (Robert E Smith Sr, AML_WATCHLIST)

  -> Entity 200004 (John Smith, AML_WATCHLIST)

  -> Entity 200010 (John Smith, AML_WATCHLIST)

2 relationship hops
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- Entity 200001 to 200004: +PHONE+EMPLOYER+SURNAME-DOB, POSSIBLY_RELATED, IS_DISCLOSED=0.

- Entity 200004 to 200010: +NAME+ADDRESS+PHONE-GENDER, POSSIBLY_SAME, IS_DISCLOSED=0.

The plus/minus feature tokens and link categories are shown as returned; both path edges are discovered relationships (IS_DISCLOSED=0), not relationships explicitly supplied in the source records. The path shows network context, not that the endpoints should be merged.

What was not found, and why

Measurement:

Only 9 of 120 resolved entities (7.5%) span multiple sources: AML_WATCHLIST/BANK_CUSTOMERS: 1, BANK_CUSTOMERS/BUSINESS_REGISTRY: 8. The separate cross-source review queue contains 15 POSSIBLY_SAME/ambiguous pairs, including 2 ambiguous pairs. For example, Entity 300003 (Maria Medina; AML_WATCHLIST:1038, DOB 12/11/73) and Entity 700030 (Maria Luis Medina Sentosa; BANK_CUSTOMERS:1036/1035/1034, DOBs 11/12/73, 11/12/73, and 11/21/73) remain an ambiguous candidate pair. Cross-source overlap is sparse, not absent.

Interpretation: The available measured evidence supports the data had little overlap to find across the resolved

cohort: only 9 resolved entities span sources, while 15 additional cross-source POSSIBLY_SAME/ambiguous pairs remain for review. This is a descriptive conclusion, not proof of source-level overlap prevalence or pipeline quality. The source-profiler uniqueness baseline and answer key are unavailable, so these counts do not establish precision, recall, F1, absolute accuracy, or pipeline underperformance.